A Compact Multiphase Divide-by-Two CMOS Frequency Divider

Yue-Fang Kuo, *Member, IEEE*, Shih-Hung Lin, Bo-Yi Chen, and Liang-Chen Liu

***Abstract*— In this article, a simple and compact eight-phase divide-by-2 frequency divider (FD) is presented and fabricated using the Taiwan Semiconductor Manufacturing Company (TSMC) 180-nm CMOS technology. The proposed divider adopted on a dual clock topology, and the alternating cross connections can effectively generate eight-phase outputs with low power consumption and few transistor counts. The proposed FD achieved a 173.3% frequency tuning range from 100 MHz to 1.4 GHz while dissipating 2.34 mW from a 1.8-V supply. The active chip size of the proposed FD core only occupies 91.7 × 52.9 μm^2.**

***Index Terms*— CMOS integrated circuits, frequency dividers (FDs), multiphase, phase-locked loops, prescaler.**

I. Introduction

THE multiphase clock generation technologies are widely used in local oscillator (LO) phase-shifting array transmitters to fine-tune output amplitude and phase of frequency modulation [1], [2], [3], [4], [5], [6]. The general frequency synthesizers adopt two-stage or four-stage *LC*-tank voltage-controlled oscillators (VCOs) to provide low-noise four-phase or eight-phase output [7], [8]. However, the multiple stages of *LC*-tank VCO require many inductors and capacitors, which occupy too much chip area and limit the operating frequency bandwidth. Furthermore, the polyphase filter (PPF) can also achieve a four-phase or eight-phase output by connecting series stages [9], [10]. Unfortunately, using too many capacitors and resistors is susceptible to process variation, resulting in a narrow bandwidth. In addition, most divide-by-two divider-based clock generators usually provide a four-phase output signal. It requires a differential input signal with two times the higher operating frequency of a divider. A four-phase input signal is necessary to realize an eight-phase output. This solution requires three divide-by-two frequency dividers (FDs) and a quadrupled differential-input signal [8], [11], [12], [13], [14]. Conversely, it can bring greater operating bandwidth. Unlike the conventional solutions mentioned above that require four-phase input, quadruple input frequency, or four-stage VCO circuitry, the proposed FD's structure with a differential input signal is simple and unnecessary for any *LC* components to achieve stable eight-phase output. This letter is organized as follows. Section II introduces the proposed FD's circuit, structure, and principle. The measurement result and conclusion are given in Sections III and IV, respectively.

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II. Circuit Design

Fig. 1 shows a traditional system block of a differential-in-quadrature-out divide-by-2 FD, whose structure is a well-known cascading *D*-latch [14]. Each *D*-latch contains two functions of sampling and holding output signal, respectively, controlled by a differential input clock (CK, CK_b). The input clock's voltage level of the first-stage $Latch_1$ is opposite to that of the second-stage $Latch_2$. When latch$_1$ performs the sampling function of $Q_{1,n+1} = D_{1b,n}$, $Latch_2$ holds the original signal of $Q_{2,n+1} = Q_{2,n}$ and vice versa. Therefore, an entire output cycle requires two input period execution times to produce a four-phase divide-by-two function.

To generate an eight-phase output signal with a division ratio of two, a duplicate cascading *D*-latch was inserted between $Latch_2$'s output (Q_2, Q_{2b}) and $Latch_1$'s input (D_1, D_{1b}) in Fig. 2. Fig. 2 shows a proposed block diagram of compact eight-phase divide-by-two FD. As observed in the upper feedback path from Q_2-to-D_3 to Q_{4b}-to-D_1, the output signal Q_2 of $Latch_2$ arrives at input signal D_1 of $Latch_1$ after two latches' delay ($2T_d$), whose phase is about $\pi/4$. Each latch schematic mainly uses seven nMOS transistors with two resistive loads, as shown in Fig. 3(a). The sampling block consists of a differential input pair (M_1, M_2) and a positive-controlled switch (M_5). A cross-coupled pair (M_3, M_4) and a negative-controlled switch (M_6) are used to hold differential output signal (Q, Q_b). The current source realized from an nMOS transistor (M_7) operated in the saturation region. Each *D*-latch executes an alternate sampling and holding signal by a differential input clock of CK and CK_b, respectively. Fig. 2 is redrawn in Fig. 4(a) based on signal transit to facilitate the analysis of the operating principle. The operating function of each latch is divided into sample (S) and hold (H) conditions, controlled by differential input signal (CK_{in}), which corresponds to the *D*-latch circuit in Fig. 3(a). $Latch_1$ and $Latch_4$ operate simultaneously in the sample mode when CK_{in} is high. Both outputs of $Latch_2$ and $Latch_3$ hold the previous voltage when CK_{in} is low. Fig. 4(b) displays the operation waveform of the proposed eight-phase divide-by-2 FD, assuming each latch's propagation delay (T_d) from *D*-to-*Q* is less than a quarter of the CK_{in} period. The positive CK_{in} is responsible for the sample mode (S_1, S_4) of $Latch_1$/$Latch_2$ and the hold mode (H_2, H_3) of $Latch_2$/$Latch_3$. Assuming each state's initial output signal (Q) is a high voltage, $Q_{4,n+1}$ samples first the previous state of $Q_{3b,n}$ under positive CK_{in}'s cycle; then, it changes from LOW to HIGH level after one T_d delay. Second, the change of $Q_{4,n+1}$ causes

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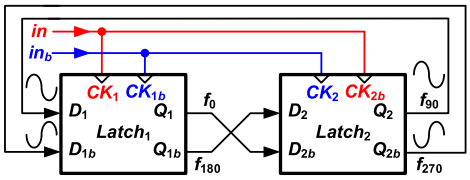


Fig. 1. Traditional block diagram of a four-phase divide-by-2 FD [14].

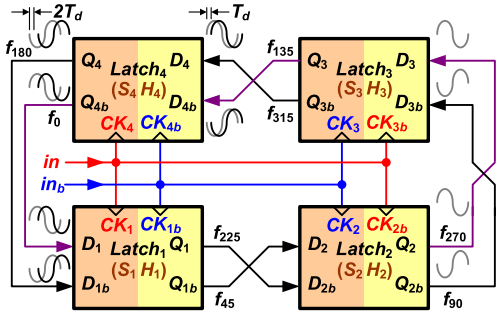


Fig. 2. Proposed system block diagram of an eight-phase divide-by-2 FD.

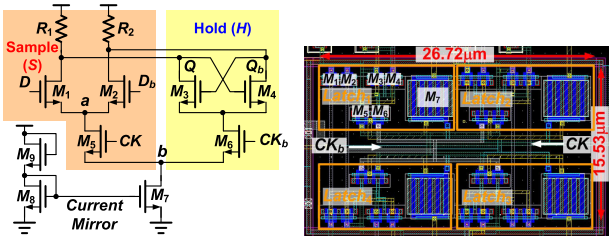
$Q_{1,n+1}$ to drop to LOW voltage. When CK_{in} transfers to a negative cycle, $Q_{4,n+1}$ presents output voltage level. Third, $Q_{2,n+1}$ starts to transfer output level from HIGH to LOW due to sample the previous state of $Q_{1,n}$. $Q_{3b,n+1}$ finally drops to LOW voltage due to $Q_{2,n+1}$'s voltage transition. Similarly, the output signal of each latch will complete one period signal after two periods of CK_{in} operation. As observed from $Q_{1,n+1}$ and $Q_{2,n+1}$, its phase difference is approximately $\pi/4$, meaning twice delay (T_d). Fig. 5(a) highlights the sample mode of the D-latch's equivalent circuit without the wire's capacitance effect. To obtain the same rise and fall times, each unit nMOS transistor except M_7 uses a channel length of 180 nm and a width of 0.9 μm. Assume that gate capacitor (C) has the same capacitance as diffusion one. The nMOS transistor's turn-on resistor value (R) is designed as twice the resistive load of R_1. Since the current source transistor of M_7 always turns on, node-b can be approximately grounded. Fig. 5(b) presents the equivalent circuit when the output voltage (Q_1) falls to the ground. Node-a has already discharged to the ground by passing through capacitance $2C$ and resistance ($R/2$). Node-Q_1 has capacitance $(6+2)C$ and resistance ($R/2$) to the ground. The Elmore delay model is introduced to calculate its time constant of transient response, which is the sum of RC product in a discharging path as follows:

$$t_{pdf} = (8C)(R/2) = 4RC. \tag{1}$$

When the input signal of D_1 changes to a LOW level, the transistor M_1 turns off. Similarly, the equivalent circuit of raising transition at node-Q_1 can be simplified in Fig. 5(c). The nMOS transistor's turn-on resistor value (R) is designed as twice the resistive load of R_1. Hence, a time constant of raising output is

$$t_{pdr} = (8C)(R_1) = (8C)(R/2) = 4RC. \tag{2}$$

As observed from (1) and (2), the rising time (t_{pdr}) is the same as the falling time (t_{pdf}) of each latch in the sampling period. The core layout size of four D-latches is 26.72×15.53 μm^2 without R_1, R_2, and output buffer,



(a) (b)

Fig. 3. (a) D-latch circuit and (b) layout and routing configuration of the proposed eight-phase divide-by-2 FD.

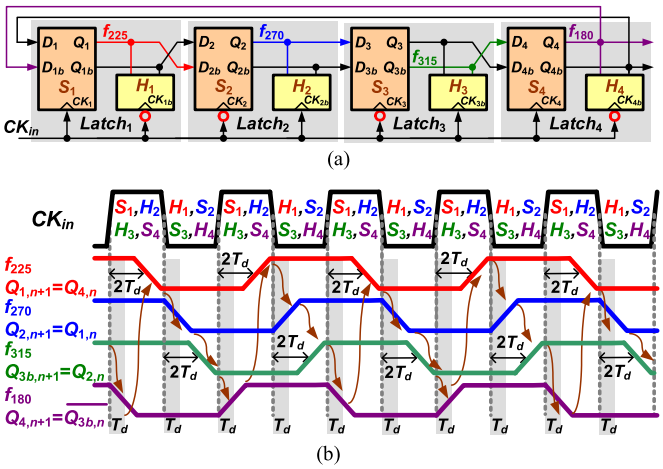


Fig. 4. (a) Sample mode and hold mode of each latch and (b) operation waveform of the proposed eight-phase divide-by-2 FD.

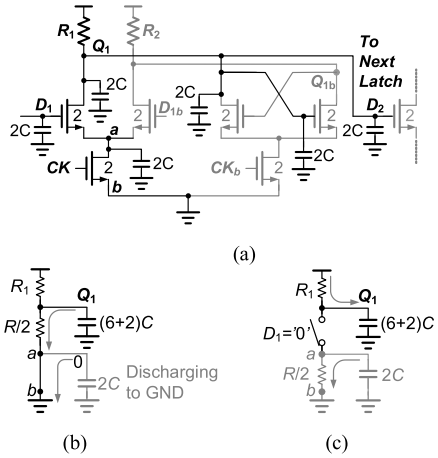


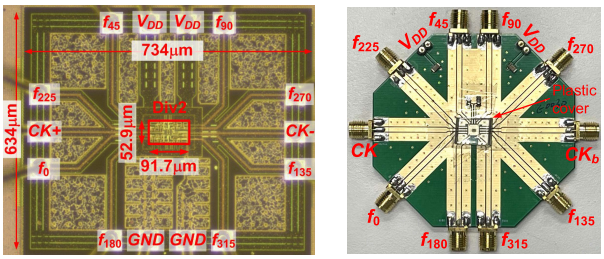
Fig. 5. (a) Equivalent circuit for D-latch, (b) falling output, and (c) rising output transition RC model.

as shown in Fig. 3(b). Each transistor's layout of the sample block (M_1, M_2), hold block (M_3, M_4), and clock block (M_5, M_6) uses two fingers to share diffusion area for reducing parasitic capacitance. The layout arrangement of the current source M_7 is adjacent to the clock block to speed up the transition time. The routing configurations of input clock signals are configured with symmetrical and equal wire lengths.

III. Measurement Results

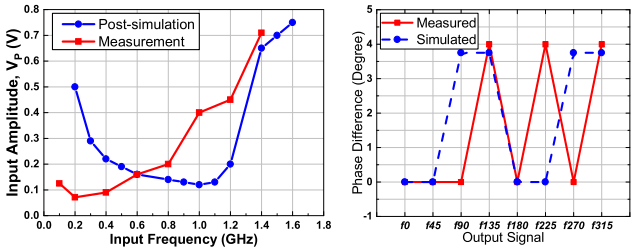
The proposed eight-phase divide-by-2 FD was designed at a 1.8-V power supply and fabricated using the Taiwan

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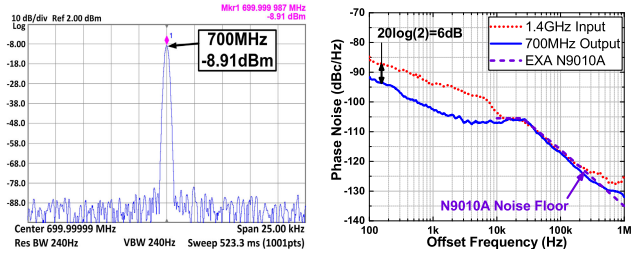
(a) (b)

Fig. 6. (a) Die photograph and (b) PCB evaluation board of the proposed FD.



(a) (b)

Fig. 7. (a) Input sensitivity characteristic and (b) output phase difference.



(a) (b)

Fig. 8. (a) Measured outputs spectrum and (b) phase noise of f_{225} at 700 MHz.

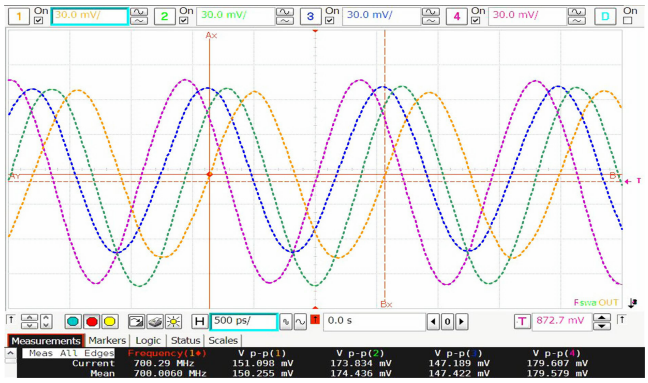


Fig. 9. Measured eight-phase outputs waveform of f_{135}, f_{90}, f_{45}, and f_0.

TABLE I

PERFORMANCE COMPARISON OF THE PUBLISHED CMOS FDS

Reference	[15]	[16]	[17]	[18]	[19]	This Work
Process (nm)	180	180	180	180	180	180
Type	CML	ILFD	ILFD	ILFD	ILFD	CML
Phases	4	2	2	2	2	8
Division	4	4	4	2	16	2
V_{dd} (V)	1.8	1.0	1.7	0.8	1.48	1.8
Freq. (GHz)	4.96–5.88	8.4–13.5	4.22–6.61	3–9	10.76–12.6	0.1–1.4
FTR*[1](%)	16.9	46.5	44.14	100	15.75	173.3
Power (mW)	4.59	9.86	8.534	9.8	12.5	2.34
Area (mm^2)	0.61[#1]	0.823	1.44	0.76	2.376	0.465
FoM_1*[2]	6.06	5.74	3.59	13.4	0.53	159.2

*[1] Frequency Tuning Range (FTR);

*[2] $FoM_1 = FTR(\%)/(\text{Power(mW)} \times \text{Area(mm}^2))$; [#1]Include VCO;

Semiconductor Manufacturing Company (TSMC) 180-nm CMOS process. Fig. 6(a) shows the proposed FD's whole die chip with dc and RF pads; its chip area was 734 × 634 μm^2. The core chip area only occupied 91.7 × 52.9 μm^2, which included R_1, R_2, and output buffer. Considering the phase error of each output signal, each signal trace has the same wire length from the output buffer to the RF pad. Since the proposed FD includes eight output signals, the chip die is directly mounted on a PCB board with bonding wires to avoid using too many GSG pads for on-wafer measurements. Fig. 6(b) demonstrates the PCB evaluation board of the proposed FD. The shape of the testing board is designed to be octagonal, so that the eight-phase microstrip lines are the same trace length. The differential input signal (CK, CK_b) was generated by a Keysight N5182B signal generator with a balun and two bias tees. An Agilent N9010A signal analyzer and a four-channel Agilent DSO9104A oscilloscope are used to measure each divide-by-2 output signal. Fig. 7(a) plots the minimum input amplitude (V_P) versus the input operating frequency of the proposed FD. The operating frequency range is from 100 MHz to 1.4 GHz. The maximum operating frequency is limited to two latches' delay ($2T_d$), which must be within half the input signal period (T_{CK}) for normal operation. The measured phase difference of each 700-MHz output signal is within 3.75°, as shown in Fig. 7(b). Fig. 8(a) shows the measured output spectrum at 700 MHz when an input clock signal is 1.4 GHz. Fig. 8(b) shows the measured phase noise comparison between a 1.4-GHz input signal and a 700-MHz divided output signal. The in-band phase noise of the divided output is 20 log(2) lower than the input signal noise. Fig. 9 shows the measured eight-phase output waveform of f_{135}, f_{90}, f_{45}, and f_0. The phase difference between adjacent outputs is approximately 45°. The mean output swing variance is about 32 mV, mainly due to the different bonding wire lengths. Table I lists the performance comparison between the proposed eight-phase divider and the various published dividers. In addition to obtaining eight-phase output, the proposed FD exhibits lower power consumption, chip size, and highest figure of merit (FoM).

IV. CONCLUSION

The dual clock topology and the alternating cross-connections are used to realize successfully the eight-phase divide-by-2 divider. The experimental results of a low power consumption and a high FTR are presented to verify the proposed FD. A simple and compact structure is suitable for integration into the multiphase sampling PLL, time-to-digital converter (TDC), clock and data recovery (CDR), or SerDes I/O applications.

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